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USING STATISTICS TO SUPPORT PROGRESSIVE SOCIAL CHANGE

WOMEN'S INVISIBILISED CHILDCARE WORK: INDIAN PATTERNS IN 2019

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We acknowledge very helpful inputs in the past on data processing and analysis from: Sonny McCann, Yizang Zhou, Prof Arkadiusz Wisniowski, Prof Amaresh Dubey, Dr. Ioana Macoveciuc, Dr Manasi Bera, and Dr Diego Perez Ruiz, among others.



OVERVIEW:

Section	Topics
Start	Introduction, definitions, and the context of challenging patriarchy whilst accepting the local norms of each area.
Middle	Description of the childcare work, the gender norms, the social groups in India 2019
Concluding part	Discussion
References	
Data Sources	
	Note: we put code chunks on open access at: https://github.com/WendyOlsen/excessworkhoursindia , A Report on India's Time-Use Survey 2019 at https://mospi.gov.in/sites/default/files/publication_reports/Report_TUS_2019_0.pdf , accessed Feb 2024.



RESEARCH QUESTION

WHAT NEW FACTORS EXPLAIN CHILDCARE HOURS PER DAY, ACROSS INDIA?

(BEYOND NUMBER-OF-CHILDREN, MARITAL STATUS, HHSIZE)

HYPOTHESES

1 SOCIAL GROUP MATTERS, BUT WE ARE NOT CLEAR ABOUT WHY THAT IS SO
(THEREFORE CONTROL FOR DEMOGRAPHICS);

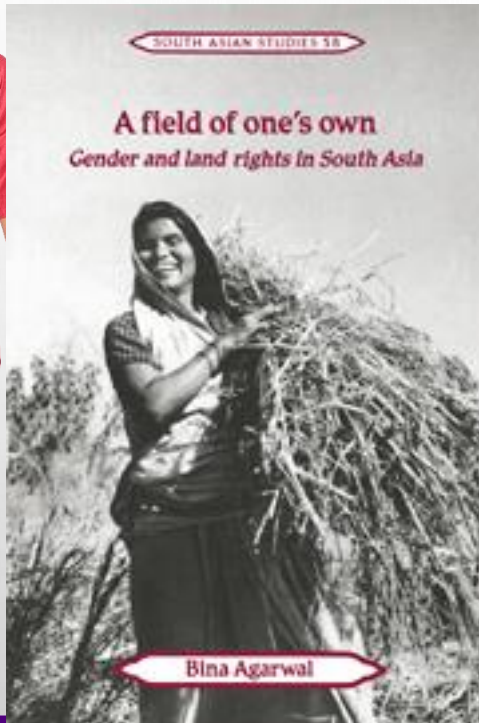
2 STATE MATTERS BECAUSE OF VARYING GENDER NORMS

**BOTH HYPOTHESES WERE FALSIFIED, AND INSTEAD,
EDUCATION AND GENDER NORMS WERE VERY IMPORTANT!**

THE CONTEXT IS, CHALLENGING PATRIARCHY WHILST ACCEPTING THE LOCAL NORMS OF EACH AREA.



ONE KEY BOOK WAS AGARWAL, A FIELD OF ONE'S OWN, 1995



How to get this book today:

Cambridge University Press

Online publication date: January 2010

Print publication year: 1995

Online ISBN: 9780511522000

DOI: <https://doi.org/10.1017/CBO9780511522000>

This book had an immense influence. Bina Agarwal found that localities and local cultures vary within each South Asian country including India.



MANCHESTER
1824

- THIS IS THE FIRST MAJOR STUDY OF GENDER AND PROPERTY IN SOUTH ASIA. IN A PIONEERING AND COMPREHENSIVE ANALYSIS BINA AGARWAL ARGUES THAT THE SINGLE MOST IMPORTANT ECONOMIC FACTOR AFFECTING WOMEN'S SITUATION IS THE GENDER GAP IN COMMAND OVER PROPERTY. IN RURAL SOUTH ASIA, THE MOST SIGNIFICANT FORM OF PROPERTY IS ARABLE LAND, A CRITICAL DETERMINANT OF ECONOMIC WELL-BEING, SOCIAL STATUS, AND EMPOWERMENT. BUT FEW WOMEN OWN LAND; FEWER CONTROL IT. DRAWING ON A VAST RANGE OF INTERDISCIPLINARY SOURCES AND HER OWN FIELD RESEARCH, AND TRACING REGIONAL VARIATIONS ACROSS FIVE COUNTRIES, THE AUTHOR INVESTIGATES THE COMPLEX BARRIERS TO WOMEN'S LAND OWNERSHIP AND CONTROL, AND HOW THEY MIGHT BE OVERCOME. THE BOOK MAKES SIGNIFICANT AND ORIGINAL CONTRIBUTIONS TO THEORY AND POLICY CONCERNING LAND REFORMS, 'BARGAINING' AND GENDER RELATIONS, WOMEN'S STATUS, AND THE NATURE OF RESISTANCE.

DEFINITIONS FIRST



What is a patriarchal standpoint? It is one which favours the control over social decisions by the elders and particularly male elders.

How we measure gender norms – use attitude and opinion questions – make a factor to smooth out the variations and gain the central tendency (Kim, et al., 2022)-WES journal

Here we use 2019 Asian Barometers - India

Sexism was embedded in questions $\leftrightarrow$ desirability bias of a patriarchal gender norm

OPERATIONALISED MEASURES



The word childcare is defined by the International Labour Organisation in its 'labour-force status' statements from the International Conference of Labour Statisticians.

When operationalising it, we now use the ICATUS, an international time-use diary category list which has 1-digit, 2-digit, and 3-digit details. We used "time spent cooking in a household with co-resident children of age 16 or below" + childcare + tutoring child(ren) +

travelling with children or family members; waiting while accompanying members...

Aggregated childcare is the total of all these.

Sex recorded as female or male; age range from 18 upward, plus spouses in agegroups 16, 17 years

Social Groups = scheduled caste people, people in scheduled tribes, Muslims not SC nor ST, other

We merged the 'NORMS' to the 'CHILDCARE' data.

WE MERGED THE ASIAN BAROMETERS DATA ON GENDER NORMS ONTO THE INDIAN TIME-USE DIARY SURVEY 2019



Data combining!



- FIRST WE CREATE SUB-GROUPS, SIMILAR TO WHEN DOING POISSON MODELLING.
- WE OPTIMISE THE MATCHING OF SUB-GROUPS BETWEEN THE TWO SURVEYS.
- WE AVOID USING SEX HERE, BECAUSE GENDER NORMS HERE DID NOT VARY BY SEX.
- WE ALSO MAY AVOID USING STATE BUT THIS IS MORE EMPIRICALLY QUESTIONABLE.
- SO, MATCH ON
- RURAL > AGE-GROUP 16-35, 36-65, 65+ > EDUCATION
- ORDINAL IN FIVE GROUPS
 - BELOW PRIMARY, PRIMARY, INCOMPLETE SECONDARY, COMPLETE SECONDARY, AND HIGHER LEVELS OF FORMAL EDUCATION
- THESE MATCH EXACTLY THE AVAILABLE MEASURES IN THE ASIAN BAROMETERS SURVEY 2019 → MERGING AT LEVEL 2.

DATASET 1 – ASIAN BAROMETERS – TABLE 1



CONFIRMATORY FACTOR,
BASED ON 3 QUESTIONS

- PREFER BOY OR GIRL, IF 1 CHILD
- SHOULD MAN ONLY DO POLITICS
- AND A THIRD LIKERT SCALE ABOUT
THE MAN'S MOTHER DOMINATING

Mean of the Gender Norm Factor (Z-Scaled) By State				
Asian Barometer Data 2019				
All respondents				
Please note we have used ordinal logit confirmatory factor analysis, giving for				
		Mean		
		gn	age	Weight factor
	Region			
	Chhattisgarh	-0.43	40.42	0.99
	Karnataka	-0.40	40.15	1.00
	Bihar	-0.33	44.94	1.02
	Uttar Pradesh	-0.23	43.33	1.02
	Haryana	-0.20	43.03	1.01
	Madhya Pradesh	-0.17	43.21	0.99
	Odisha	-0.16	45.55	1.06
	Tamil Nadu	-0.04	40.01	1.02
	Jharkhand	-0.03	40.55	1.01
	Gujarat	-0.01	44.88	1.05
	West Bengal	0.07	43.93	1.11
	Rajasthan	0.14	39.88	0.97
	Maharashtra	0.19	44.13	1.04
	Kerala	0.29	48.21	1.07
	Telangana	0.36	36.74	0.97
	Andhra Pradesh	0.38	42.41	1.03
	Assam	0.66	40.87	1.01
	Punjab	0.67	45.04	0.91
	Delhi	0.85	40.61	1.09
	Total	0.00	42.76	1.03



Boy-
preferring;
man only in
politics etc.

Not Boy-
preferring;
NOT man only
in politics etc.

FINDINGS – ASIAN BAROMETERS – TABLE 2



Mean of the Gender Norm Factor and Case-Weights by Education

Asian Barometer Data 2019

All respondents

Please note the gender norm index ran from -3 to +3 approximately



			Mean		Standard deviation		Number of nonmissing values	
			Gender norm	Weight factor	Gender norm	Weight factor	Gender norm	Weight factor
		edu						
		Below Primary	-0.13	1.08	0.80	0.18	1609.00	1609.00
		Primary	-0.04	1.03	0.81	0.13	547.00	547.00
		Incomplete Secondary	0.09	1.03	0.83	0.27	624.00	624.00
		Complete Secondary	0.07	0.99	0.85	0.16	1381.00	1381.00
		Higher Educ	0.08	0.97	0.92	0.25	1157.00	1157.00
		Total	0.00	1.03	0.85	0.21	5318.00	5318.00

DATASET 2 – INDIAN TIME-USE SURVEY 2019



See also our paper in *Development & Practice* 2023
(extreme working hours of child labourers)

TABLE 2

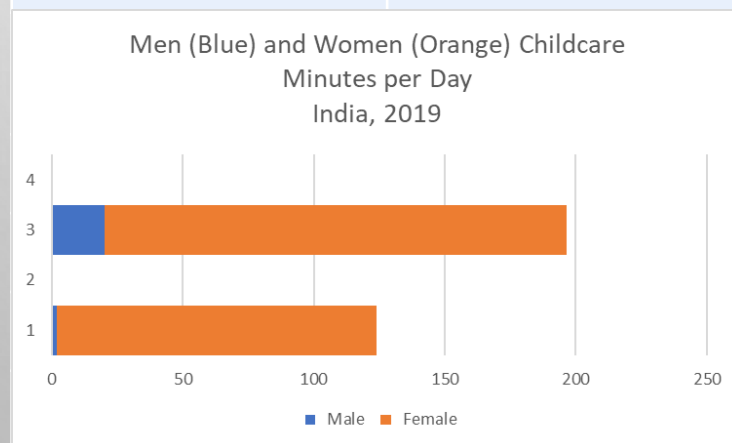
Mean of the Cooking Variable and the Overall Total Childcare Time
India Time-Use 2019

All adult respondents

Please note the adults were aged 18+, or were married age 16 or 17 years.

TABLE 3

		Mean (Minutes On Recall Day)	
		Cooking time	All child care time
Male		2.0	20.1
Female		121.9	176.3
Total		61.9	98.1



FINDINGS – INDIAN TIME-USE SURVEY 2019

See also our paper in *Development & Practice* 2023
(extreme working hours of child labourers)



Mean of the cooking variable and the overall total childcare time				TABLE 3 - details			
India Time-Use 2019							
All adult respondents							
Please note the adults were aged 18+, or were married age 16 or 17 years.							
		Mean (Minutes)		Standard deviation		Count	
		Cook time	All child care time	Cook time	All child care time	Cook time	All child care time
	Male	2.0	20.1	18.7	50.9	179093	179093
	Female	121.9	176.3	141.9	194.5	178895	178895 ¹¹
	Total	61.9	98.1	117.6	162.2	357988	357988

FINDINGS: REGARDING WHETHER THE GENDER NORMS AFFECTED THE CHILDCARE WORK TIME



- NO SIGNIFICANT DIFFERENCE ACROSS SOCIAL GROUPS

socialgroup2					
Others	ST	SC	OBC	Muslim Non SC/ST	Total

- DIFFERENCES BY STATE ARE VERY SMALL, JUST A FEW MINUTES OF WORK
- BUT DIFFERENCES BY 'GENDER NORMS INDEX' AT THE SMALL GROUP LEVEL ARE LARGE.
- 1 UNIT RISE IN GN INDEX IS ASSOCIATED WITH 66 MINUTES MORE!
- THIS IS WHERE A CULTURAL GROUP NORM IS ASSOCIATED WITH WOMEN DOING MORE CHILDCARE IN TERMS OF TIME SPENT ON THE VARIOUS ACTIVITIES, INCLUDING COOKING WITH CHILDREN AROUND.

FINDINGS 2

REGARDING WHETHER THE GENDER NORMS AFFECTED THE CHILDCARE WORK TIME AFTER ALLOWING FOR STATE LOCATION



- CRITICAL ASSESSMENT OF ASIAN BAROMETERS SAMPLING METHODS

- GOOD TREATMENT OF DROPOUTS, NON-REPLACEMENT
- HAD A HIGH SAMPLING FRACTION TO ALLOW FOR NON-REPLACED DROPOUTS
- THE VERY LONG INTERVIEW HAD A LOT OF OLDER PEOPLE RESPONDING RELATIVE TO YOUNG.
 - INTRODUCED WEIGHTS WITHIN ASIAN BAROMETERS RELEASED SURVEY DATA
 - THEY TESTED THESE WEIGHTS CAREFULLY
 - SO YOUNG PEOPLE GOT HIGHER WEIGHT EACH

- LASTLY, THE POOR CHOICE OF A GEOGRAPHIC CLUSTER SAMPLING METHOD WITHIN STATE HAS WEAKENED THE GENERALITY AND REPRESENTATIVENESS (OFTEN, 2 TO 4 ASSEMBLY CONSTITUENCIES)

FINDINGS 2: TABLE 4.

THE GENDER NORMS AFFECTED CHILDCARE WORK TIME AFTER ALLOWING FOR STATE LOCATION? YES. FEMALE = +301 MINUTES. + IF **GROUP GENDER NORMS PROGRESSIVE = + > 55 MINUTES AVERAGE!**

STATE EFFECTS WERE DISCERNED ONCE THE GENDER NORMS WERE ALLOWED FOR. A FEW OF THE STATE EFFECTS ARE LARGE.



EXAMPLES:		MEASUREMENT IN MINUTES PER DAY OF CHILDCARE					
		Coefficient	Std. error	T	P(t)	Lower Bound	Upper Bound
Rajasthan		-37.9	5.7	-6.7	0	-49.1	-26.7
Haryana		-27.5	6.1	-4.5	0	-39.5	-15.6
Delhi		-25.0	7.0	-3.6	0	-38.7	-11.4
Uttar Pradesh		-21.8	5.5	-3.9	0	-32.7	-11.0
Jharkhand		38.1	6.1	6.2	0	26.1	50.1
Uttarakhand		38.2	8.4	4.5	0	21.6	54.7
Odisha		39.6	6.3	6.3	0	27.2	52.0
West Bengal		54.5	5.8	9.4	0	43.2	65.8
Bihar		59.8	5.7	10.6	0	48.7	70.9
Assam		81.8	5.8	14.1	0	70.4	93.2

The base case state was Andhra Pradesh and controls were social group, rural, has someone

in a teaching or computing profession in the household, number of children, agegroup, gender.



THE REGRESSION RESULTS (TESTS)



	Sample:	Overall Test:		Pseudo R-squared is good.
Tobit regression Number of obs = 357,988	Uncensored = 146,163 Limits: Lower = 0 Left-censored = 211,825 (Zero Childcare)	$F(48, 357940) = 2722.73$ $\text{Prob} > F = 0.0000$	$\text{Log pseudolikelihood} = -1262639.4$	$\text{Pseudo } R^2 = 0.1011$

DISCUSSION

Women do most of the childcare

Certain kinds of women do the most

**Education is a protective factor against widespread pro-male sexism.
But Progressive Norms are also associated with high levels of hours-per-week of childcare work**

Gender norms vary by education, not by state

**Childcare work time does vary by state,
And even more so by local group gender norms
(Progressive = MORE work)**

**As a result, one should treat India like European Union,
not like a small island state**

(Large spatial residuals)

Scientific method

Has had 3 key hypotheses.

We tested each one.

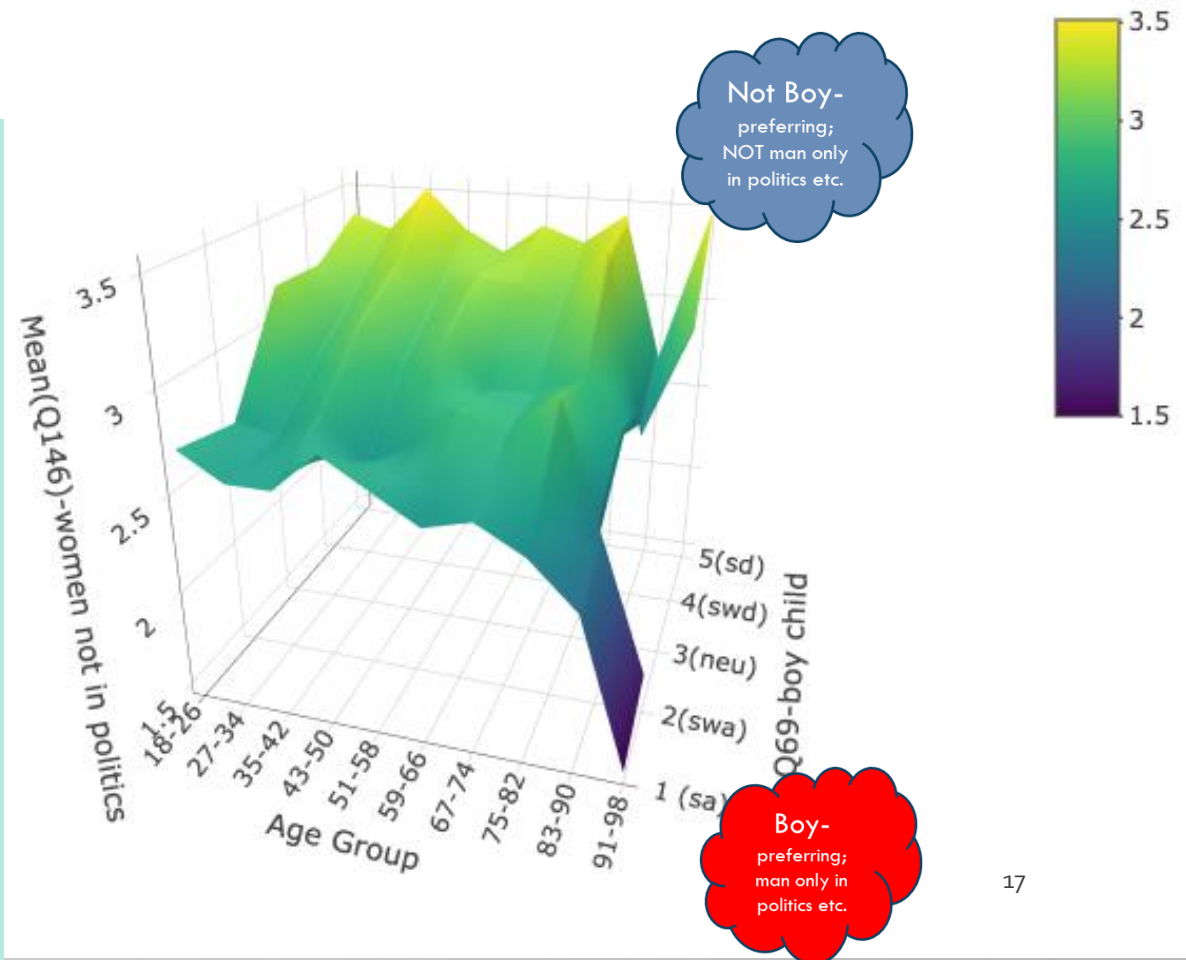
We were surprised sometimes by the findings, eg in the background, non-difference of gender norms by sex contrasts with naïve expectations.

GRAPHICAL APPROACH TO THE FINDINGS

ASIAN BAROMETERS SHOWN⇒

GENDER NORMS HIGHLY CORRELATED ↓↓

	Robust					
allchildcaretime	Coefficient	std. err.	t	P> t	[95% conf. interval]	
-----+-----						
gender	301.1068	1.076889	279.61	0.000	298.9961	303.2174
agegroup						
2	-.75.91933	.9860008	-77.00	0.000	-77.85186	-73.98679
3	-143.7858	2.78459	-51.64	0.000	-149.2435	-138.3281
numchild	225.0759	2.315317	97.21	0.000	220.538	229.6139
numchild2	-30.32982	.64938	-46.71	0.000	-31.60258	-29.05705
numadult	-13.97016	.34863	-40.07	0.000	-14.65347	-13.28686
hasteach	6.698178	4.288748	1.56	0.118	-1.707642	15.104
hascompoccup	16.17882	3.158329	5.12	0.000	9.988592	22.36906
sc	-4.82571	1.272034	-3.79	0.000	-7.31886	-2.332559
st	-12.0681	1.625505	-7.42	0.000	-15.25404	-8.882157
muslim	-3.599906	1.561779	-2.31	0.021	-6.660948	-.5388646
gngroupmean	55.57491	4.922029	11.29	0.000	45.92788	65.22194
rural	3.363136	1.548991	2.17	0.030	-.3271591	6.399113
state						



BEWARE OF ISSUES WITH THE INDIAN DATA



INDIAN TIME-USE SURVEY 2019

- HAS HAD FEW USERS RELATIVE TO ITS IMPORTANCE
- WAS RATHER DIFFICULT TO USE INITIALLY
 - CATEGORY CODES ARE NOT IN THE DOCUMENTATION
 - (USE THE ITUS REPORT FOR THESE)
 - CAN LOOK UP ICATUS (2016) DOCUMENT
 - THESE CODES CHANGE FROM TIME TO TIME!
- IT HAD A GOOD SAMPLING METHOD

THE HOURS WORKED IN THE ONE RECALL DAY WERE MEASURED OUT IN 30-MINUTE BLOCS

- 30 MINUTE BLOCKS GIVE A LARGE ERROR BAR
- HIDDEN MEASUREMENT ERROR SUCH AS MASKING ALL THE <15 MINUTE ACTIVITIES LIKE BRUSHING THE CHILD'S TEETH
 - CHANGING A DIAPER
- LIKELY TO LEAD TO UNDERESTIMATING CHILDCARE TIME



SUMMARY AND POINTERS FORWARD

SURVEY OF THE MAIN POINTS TODAY

- DATA COMBINING CAN INCREASE THE COST-EFFECTIVENESS OF THE TIME-USE SURVEY
- GENDER NORMS

POINTERS TO HOW TIME-USE DATA WILL BE ORGANISED IN FUTURE, IN MANY COUNTRIES

- MULTI STAGE RANDOM SAMPLING
- CLUSTER SAMPLING STAGE IS A RISK ZONE FOR NON-REPRESENTATIVENESS
- COSTS RISE, SO STATISTICIANS CAN ARGUE THE **CASE FOR TIMEUSE SPENDING** IF...
 - A THEY GET IN THE ROOM
 - B THEY CAN SHOW BENEFITS FROM THE DATA
 - C THEY SAY CLEARLY WHO BENEFITS
 - D POLICY-PRACTITIONERS DIRECTLY, AND CHILDCARE WORKERS AND FATHERS INDIRECTLY, BENEFIT FROM THE INFO.





SOME REFERENCES

INDIAN DATA AND EARLY STUDIES

S. Rao and Vijayamba R., online document on "Using the India TUS Unit Level Data: Some notes on method", URL <https://fas.org.in/using-the-india-tus-unit-level-data-some-notes-on-method-and-some-stata-code/>, accessed March 2022.

Full Open Access Methods:

github.com/WendyOlsen/combining-cmie-itus-data

Dev't In Practice Paper Methods:

github.com/WendyOlsen/excessworkhoursindi
a

EMPIRICAL RESEARCH STUDIES

KIM, JIHYE, AND WENDY OLSEN (2023), HARMFUL CHILD LABOUR IN INDIA FROM A TIME-USE PERSPECTIVE, DEVELOPMENT IN PRACTICE, 33:2, DOI [HTTPS://WWW.TANDFONLINE.COM/DOI/FULL/10.1080/09614524.2022.2155620](https://www.tandfonline.com/doi/full/10.1080/09614524.2022.2155620), OPEN ACCESS.





FURTHER INFORMATION SOURCES

- *FOR ICATUS CODES USED AT 1-DIGIT LEVEL, SEE UN (2021), INTERNATIONAL CLASSIFICATION OF ACTIVITIES FOR TIME-USE STATISTICS 2016, DEPARTMENT OF ECONOMIC AND SOCIAL AFFAIRS STATISTICS DIVISION, STATISTICAL PAPERS SERIES M NO. 98, NEW YORK: UNITED NATIONS. URL [HTTPS://UNSTATS.UN.ORG/UNSD/GENDER/TIMEUSE/23012019%20ICATUS.PDF](https://unstats.un.org/unsd/gender/timeuse/23012019%20ICATUS.PDF)
- SPECIFIC CODES ARE IN THE REPORT PUBLISHED 2021
- INDIAN TIME USE SURVEY (ITUS) MICRODATA 2019 (NSS, 2020) AVAILABLE AT:
*[HTTPS://MOSPI.GOV.IN/WEB/MOSPI/DOWNLOAD-TABLES-DATA/-/REPORTS/VIEW/TEMPLATETWO/20702?Q=TBDCAT](https://mospi.gov.in/web/mospi/download-tables-data/-/reports/view/templatetwo/20702?q=TBDCAT). ACCESSED SEPT. 15, 2022.
- THERE IS A LOGIN STAGE AT MOSPI WEBSITE.
 - TEXT DATA LAYOUT FOR TIME USE SURVEY : SCHEDULE-10.6, URL [HTTPS://MOSPI.GOV.IN/SITES/DEFAULT/FILES/README_TUS106.PDF](https://mospi.gov.in/sites/default/files/readme_tus106.pdf), AVAILABLE VIA [HTTPS://MOSPI.GOV.IN/TIME-USE-SURVEY-0](https://mospi.gov.in/time-use-survey-0)
- NIC INFO IS AT:
 - [HTTPS://WWW.NCS.GOV.IN/DOCUMENTS/NIC_SECTOR.PDF](https://www.ncs.gov.in/documents/nic_sector.pdf)

